

Gold Price Prediction Using Python and Machine Learning

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➤ Dataset Overview:-

This is the dataset I used for the Gold Price Prediction project: **609 daily records** across **10 fields** — Date, Day_of_Week, Day_of_Week_Num, **24K_PKR_per_Tola**, 24K_Daily_Change_PKR, 24K_Daily_Change_Pct, **Is_Observed**, Fill_Basis_Date, **XAU_USD**, and **USD_PKR**, covering the period from **January 2025 to September 2026**. When I checked the dataset, there were **no duplicate rows** and **no missing values** at all, since the price columns had already been **forward-filled** for non-trading days and the **Is_Observed** column was used to flag which rows were genuine recorded prices versus filled-in ones. The target column, **24K_PKR_per_Tola**, had **223 unique price values**, while the two external columns I merged in, **XAU_USD** and **USD_PKR**, came from separate sources and were filled **forward and backward** to cover a small number of gaps at the start and end of the series.

➤ Data Pre-processing:-

First, the two external datasets — **XAU_USD** and **USD_PKR** — were merged into the main dataframe using a **left join on Date**. Since some dates didn't have matching values right away, missing entries were filled using **forward-fill (ffill)**, and any values still missing after that were filled using **backward-fill (bfill)**. Next, the dataset was filtered to keep only rows where **Is_Observed == 1**, removing the forward-filled "fake" days so only genuine recorded prices remained, and the data was sorted by **Date** to keep the correct time order. Then, several features were engineered from the price column: **lag_1** (yesterday's price), **rolling_mean_7** and **rolling_mean_30** (7-day and 30-day average price), **rolling_std_7** and **rolling_std_30** (7-day and 30-day price volatility), **day_index** (a simple day counter), and **momentum_7** (percentage price change over 7 days). Similar lag, percentage-change, and rolling-mean features were also created for **XAU_USD** and **USD_PKR** individually. A new column called **implied_price** was calculated using XAU_USD and USD_PKR converted from USD/ounce into PKR/tola, representing what the local price *should* theoretically be, along with its own **implied_price_lag_1** and **implied_change_pct**. A **price_vs_implied_gap** column (and its lagged version) was created by subtracting implied_price from the actual price, to see how far local prices moved from their theoretical fair value. After creating all these features, rows with **NaN** values (caused by the rolling windows needing prior days) were dropped. Finally, a **recency_weight** column was added using **exponential decay** with a 60-day half-life, so more recent rows would carry more importance during model training.

➤ Visualization Display & Insights

○ Price Trend (Line Chart):-



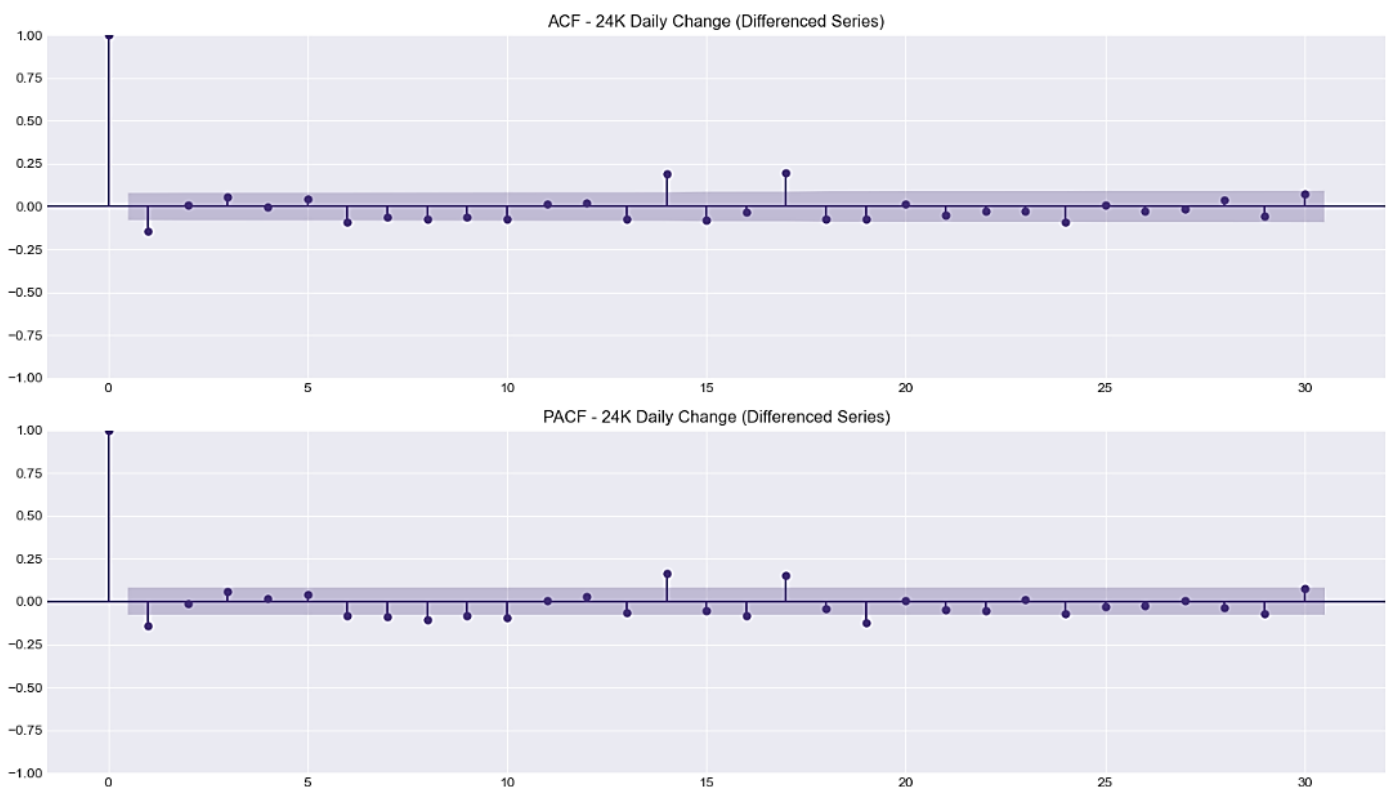
- ✓ The chart shows a steady, predictable uptrend from **290,000 to 430,000 PKR** between January 2025 and May 2026, followed by a sharp shift into high volatility from **June 2026** onward, with sudden jumps and drops replacing the smooth climb. It peaks near **485,000 PKR** close to September 2026 before pulling back to around **450,000 PKR**. The market clearly entered a much more unstable phase in the last few months, which means a model trained equally on the full history would underestimate recent volatility recent data needs to carry more weight for accurate predictions.

○ Price vs Rolling Mean & 30-Day Rolling std Dev (Line Chart):-



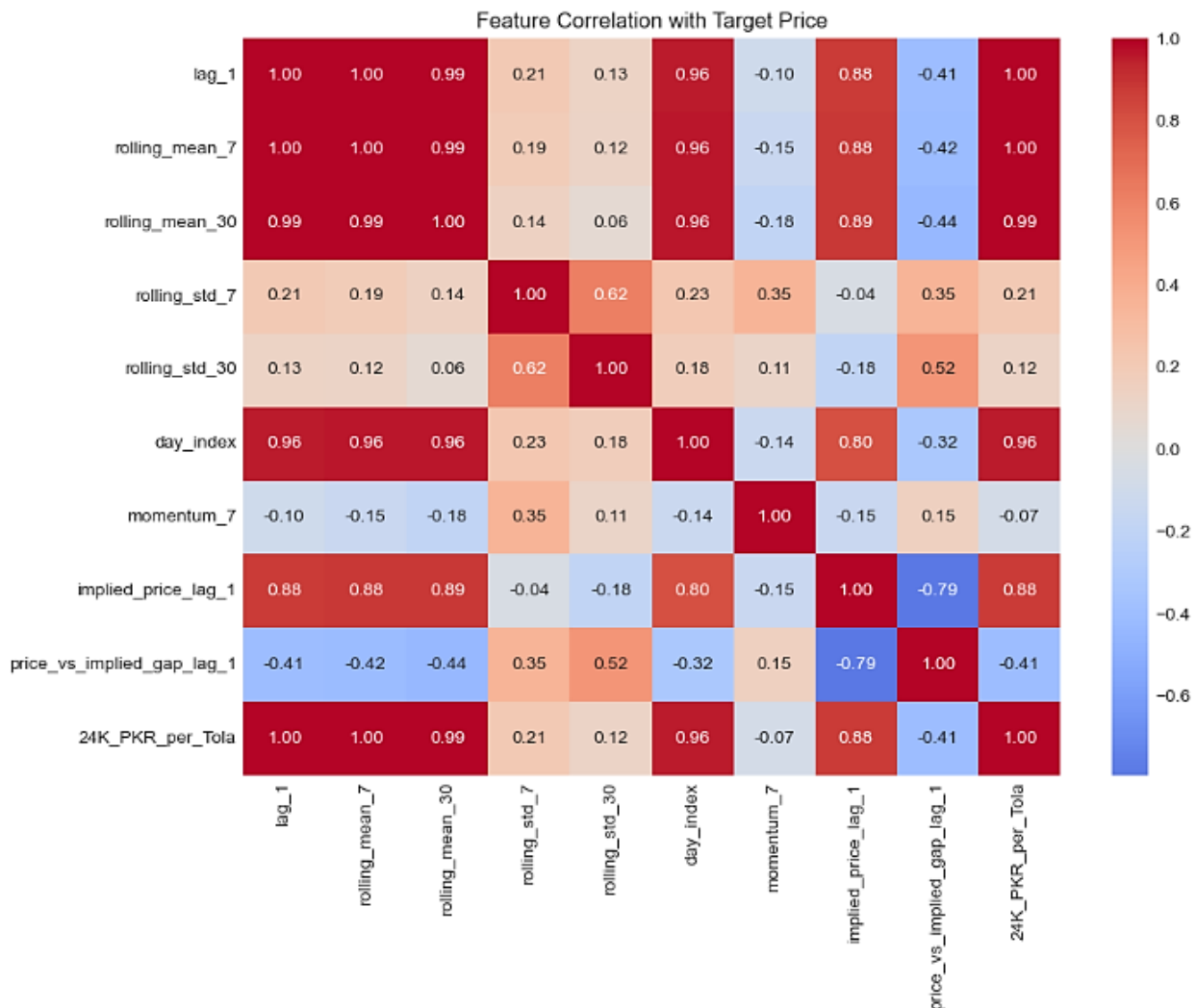
- ✓ The top chart compares the actual price against its **30-day rolling mean**, and from January 2025 to around **May 2026**, the two lines stay close together, with only small spikes breaking away from the smooth blue trend line, showing a stable market. From **June 2026** onward, the actual price line pulls sharply away from the rolling mean, swinging above and below it instead of following it closely. The bottom chart, showing **30-day rolling standard deviation** (volatility), confirms this exact same shift — std dev stays low and controlled, mostly under **5,000**, for almost the entire dataset, then jumps sharply to **13,000-14,000** starting **June 2026**, dips briefly, and spikes again to nearly **20,000** by September 2026.
- ✓ **Finding:** both charts confirm the same thing from two different angles — the market entered a clearly more unstable phase in the last three months, so the model needs to expect wider swings rather than a smooth continuation of the earlier steady uptrend.

○ ACF & PACF Plots:-



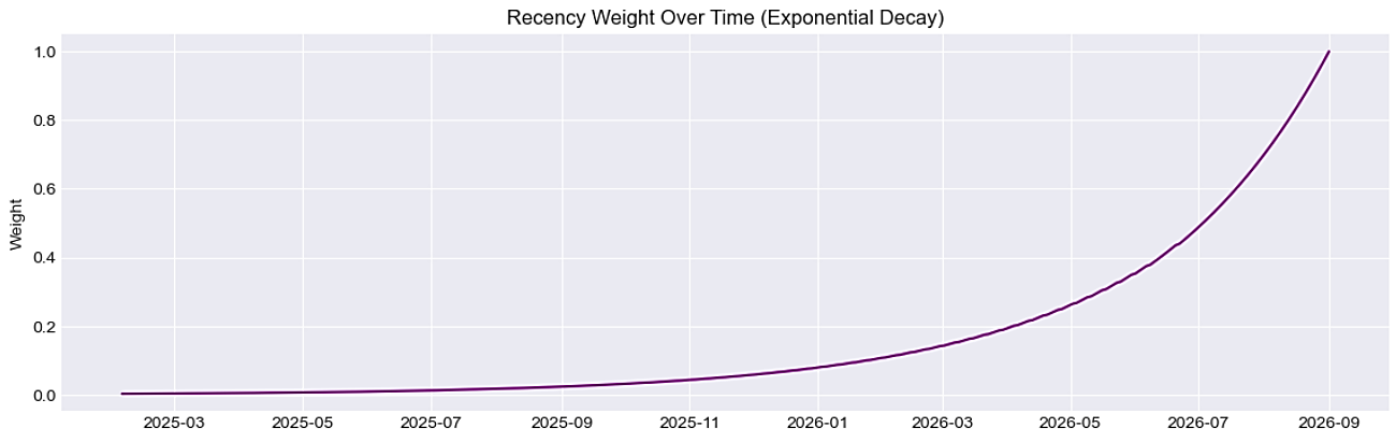
- ✓ Lag 0 is always **1.0**, which just confirms the series matches itself, not a real finding. **Lag 1** sits clearly outside the blue confidence band with a negative value (around **-0.15**) in both ACF and PACF, making it the only strong, consistent signal in the data. **Lag 14** and **Lag 17** is up slightly outside the band (around **0.15-0.18**), hinting at a weak, possibly biweekly pattern, while every other lag stays inside the band with no meaningful relationship.
- ✓ **Finding:** daily gold price changes behave close to random noise only lag_1 gives a dependable signal, so rolling averages and trend-based features will matter more than deep lag features for the model.

○ Feature Correlation with Target Price (Heatmap):-



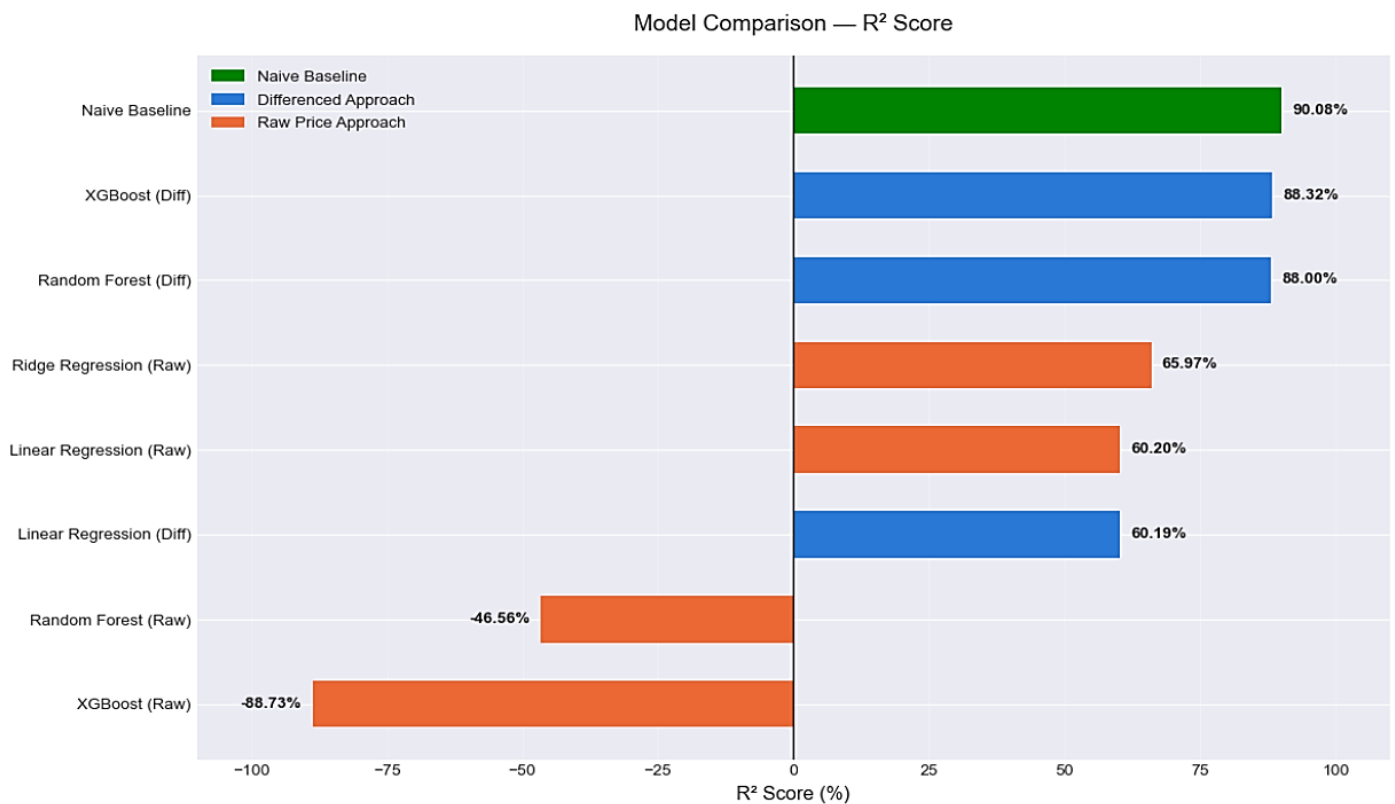
- ✓ **lag_1, rolling_mean_7, rolling_mean_30, and day_index** all show near-perfect correlation with the target price (**0.96 to 1.00**), which makes sense since gold price has a strong overall uptrend. These same four features are also highly correlated with each other (**0.96-1.00**), meaning they're carrying almost the same information and are largely redundant. **rolling_std_7** and **rolling_std_30** have much weaker correlation with price (**0.21** and **0.12**), but they still add useful volatility information the trend features don't capture.
- ✓ **Finding: implied_price_lag_1** stands out with a strong correlation (**0.88**) despite not overlapping with the trend features, making it a genuinely useful, non-redundant signal while **momentum_7** (**-0.07**) adds almost nothing on its own.

○ Recency Weight Over Time (Line Chart):-



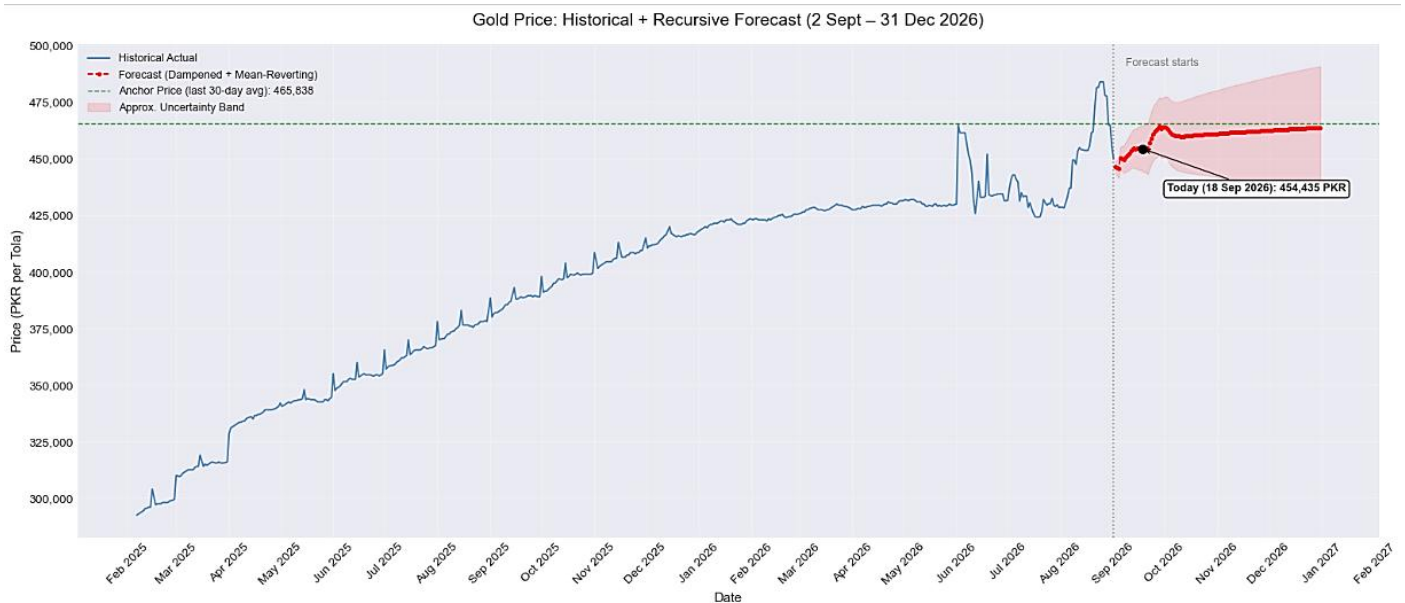
- ✓ The weight stays very close to **0** for most of the dataset, from **February 2025 to around November 2025**, then starts rising slowly before climbing sharply in the last few months, reaching exactly **1.0** at the most recent date (**September 2026**).
- ✓ **Finding:** this confirms the recency weighting is working correctly older dates get almost no importance while the most recent dates get the highest importance, matching the decision to give more weight to recent data for better prediction accuracy.

○ Models Accuracy Comparison (Bar Chart):-



- ✓ The **Naive Baseline** scores highest at **90.08%**, followed closely by the **differenced approach** models, **XGBoost (88.32%)** and **Random Forest (88.00%)**, shown in blue. The **raw price** models, shown in orange, perform much worse — **Ridge (65.97%)** and **Linear Regression (60.20%)** land in the middle, while **Random Forest (Raw)** and **XGBoost (Raw)** drop into negative territory (**-46.56%** and **-88.73%**), meaning they perform worse than just guessing the average.
- ✓ **Finding:** switching from predicting raw price to predicting the daily *change* in price turned XGBoost and Random Forest from the worst-performing models into the best real ML models, almost matching the naive baseline.

○ Overall Prediction (Line Chart):-



- ✓ After the sharp climb to nearly **485,000 PKR**, the forecast (red) pulls back and stabilizes close to the **anchor price of 465,838 PKR**, with today's (18 Sept 2026) predicted value at **454,435 PKR**. The **uncertainty band** (pink shading) widens as the forecast moves further into the future, reflecting growing prediction risk.
- ✓ **Finding:** the mean-reversion and dampening keep the forecast anchored near recent levels instead of following the sharp swings seen just before the forecast start.

➤ Conclusion:-

The XGBoost (Differenced) model predicts gold prices reasonably well, reaching an **R² score of 88.32%** and a **MAPE of just 0.96%**, very close to the naive baseline and the best result among all models tested. Still, the predictions can't be perfectly accurate, since gold prices are shaped by real-world factors the model can't see global drivers like Fed rates and geopolitics, and local drivers like SBP policy and seasonal demand. Reliability also drops further into the forecast, since each day's prediction feeds off the previous day's guess rather than real data, letting small errors

accumulate over time. Even so, the dampening and mean-reversion keep the forecast anchored near actual recent prices, and validation on known dates (**1.08% MAPE**) confirms it holds up consistently rather than by chance. Overall, the model captures the underlying trend well, but sudden market shocks remain fundamentally unpredictable so the forecast is best read as a reasonable estimate, not a guarantee.